

## 练习十五 统计物理学基础（三）

1.  $2\alpha xL$

2.  $2.19 \times 10^8 (J)$

3. (4); 4. (2)

5.  $P_0 L_0 = (P_0 + \frac{4\alpha}{d})h \quad h=19.72\text{mm}$

6. 在大气压  $P_0 = 1.0136 \times 10^5 \text{Pa}$  时，泡内压强  $P = P_0 + \frac{4\alpha}{R_1}$ ，移到气压为  $P'_0$  时泡内

$$P' = P'_0 + \frac{4\alpha}{R_2} \quad \therefore P \times \frac{4}{3} \pi R_1^3 = P' \cdot \frac{4}{3} \pi R_2^3 \quad \left( P_0 + \frac{4\alpha}{R_1} \right) \cdot R_1^3 = \left( P'_0 + \frac{4\alpha}{R_2} \right) R_2^3$$

$$P'_0 = \left( P_0 + \frac{4\alpha}{R_1} \right) \left( \frac{R_1}{R_2} \right)^3 - \frac{4\alpha}{R_2} = 12672.5 (\text{Pa})$$